

CRC (Cyclic Redundancy Check).

$n \Rightarrow 101100110 \Rightarrow \text{Divident}$

$m \Rightarrow 1001$

⇒ Divisor.

$$|m| = 4 - 1 = 3$$

$m \Rightarrow 1001 \Rightarrow 11110$
A (sender side) $\xrightarrow{\text{Data}}$ B (Receiver side)
is there any error or not?

(4) $|m| - 1$ 0's will be appended at the end of n .
 Replace by CRC

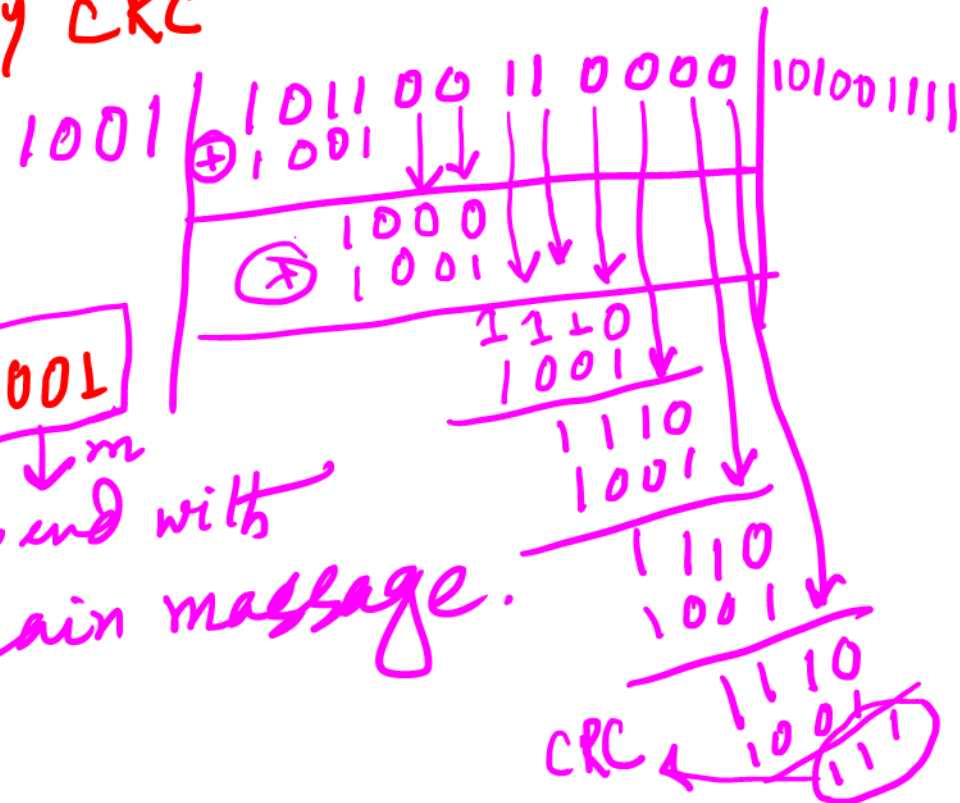
(ii) $101100110000 \Rightarrow n$

(iii) $n \div m$ (Division).

(iv) 101100110111

(v) sent data! →

(iv) Sent Data $\rightarrow$ 101100110111 **1001** | 1
 ↓
 append with
 main message.



Receiver Side-

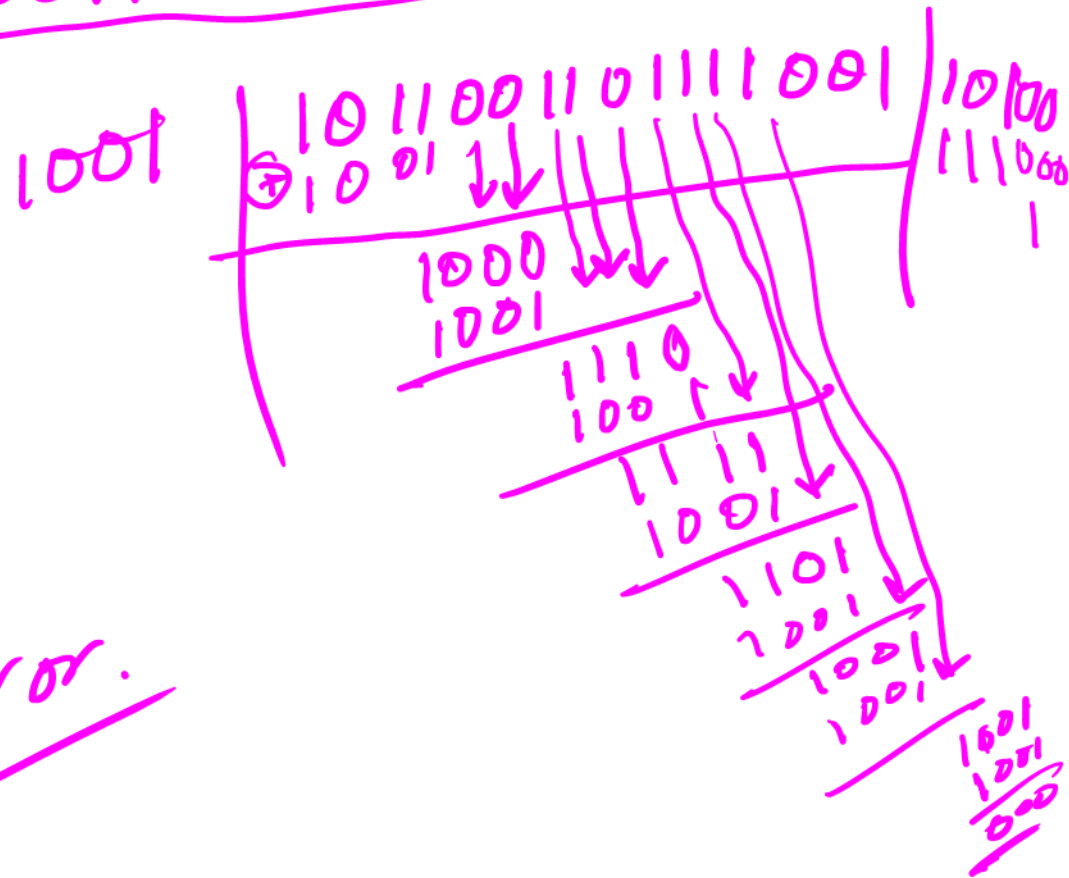
(i) Received Data $\Rightarrow$ 101100110111001 $\rightarrow$

(ii) $m \Rightarrow 1001$

(iii) Division.

(iv) If CRC is zero then
no error.

(v) If CRC is non-zero
then will be some error.



Ex! → Data $\Rightarrow x^7 + x^5 + x^2 + x + 1 \Rightarrow 10100111$
 Divisor $\Rightarrow x^3 + x + 1 \Rightarrow \underline{1011}$
 $1x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x + 1$

Find CRC Value!

→ Data $\Rightarrow 10100111\underline{000} \Rightarrow n \cdot m$ $m = 1011$

Remainder $\Rightarrow 11$

Polynomial $\Rightarrow x + 1$

$\therefore \text{CRC} \Rightarrow \underline{f(x)} = \underline{x+1}$

$$\begin{array}{r} 010 \\ 11 \\ \hline 011 \end{array}$$

Check sum

Data-1 $\Rightarrow$ $\begin{matrix} 2 & 3 & 3 & 1 & 2 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{matrix} \rightarrow 1$
 $\rightarrow 2$

Data-2 $\Rightarrow$ $\begin{matrix} 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 \end{matrix} \rightarrow 3$

Data-3 $\Rightarrow$ $\begin{matrix} 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 \end{matrix} \rightarrow 4$

Data-4 $\Rightarrow$ $\begin{matrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{matrix} \rightarrow 5$

Data-5 $\Rightarrow$ $\begin{matrix} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{matrix}$

$\begin{matrix} 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \end{matrix}$
 Discant. $\downarrow$ 1's comp $\Rightarrow C$ (checker)
 i g/vior. $\downarrow$ $\begin{matrix} 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{matrix}$

Data sent $\Rightarrow$

1	2	3	4	5	C
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Receiverside !,

Data-1 →
Data-2 →
Data-3 →
Data-4 →
Data-5 →
check+c →

if the result is all 1;
then there is no error.
Otherwise - error.

Step-

(i) sum

(ii) 1's Complement

(iii) all Data + checker - sent

(i) Receive
(ii) sum.
(iii)

